Nebula-Net: Compact Vision Encoders Revisited

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Abstract

We present Nebula-Net, a compact vision encoder that combines cross-modal attention with a lightweight convolutional trunk. Nebula-Net is trained on a mixture of natural images and synthetic augmentations, and evaluated on four standard benchmarks. On ImageNet the model reaches 78.4% top-1 accuracy with only 12.3M parameters, outperforming prior art at comparable compute.

1. Introduction

Efficient image classification remains a central problem in computer vision. While large transformers have set records on accuracy, their parameter counts and latency profiles make them impractical for on-device deployment. In this work we revisit small convolutional architectures and show that a careful pairing of depth-wise separable convolutions with lightweight cross-attention layers yields a strong accuracy/compute trade-off.

Our contributions are threefold. First, we introduce Nebula-Net, a 12.3M-parameter backbone. Second, we propose a training recipe that combines RandAugment-style augmentations with a label-smoothed cross-entropy loss. Third, we provide extensive ablations across four benchmarks.

2. Method

Nebula-Net consists of five stages. Each stage halves the spatial resolution and doubles the channel count. Within each stage, inverted residual blocks are interleaved with a single cross-attention layer that exchanges information across spatial tokens. We use GELU activations throughout and apply dropout with probability 0.1 on the final classifier layer.

We train for 120 epochs using AdamW with a base learning rate of 3e-4 and cosine decay. Batch size is 512. Mixed-precision training is used throughout.

3. Experiments

We evaluate Nebula-Net on CIFAR-10, CIFAR-100, ImageNet, and SVHN. All datasets are used at their standard resolutions. We report top-1 accuracy on the held-out validation split for each benchmark. The complete results are shown in Table 1 and visualized in Figure 2.

Table 1 also reports the approximate inference latency measured on a single consumer GPU with batch size 1 at FP16 precision.

Dataset	Accuracy (%)	Params (M)	Latency (ms)	Year
CIFAR-10	94.2	12.3	4.1	2026
CIFAR-100	76.8	12.3	4.1	2026
ImageNet	78.4	12.3	6.8	2026
SVHN	96.1	12.3	3.9	2026

Table 1: Results on four benchmarks.

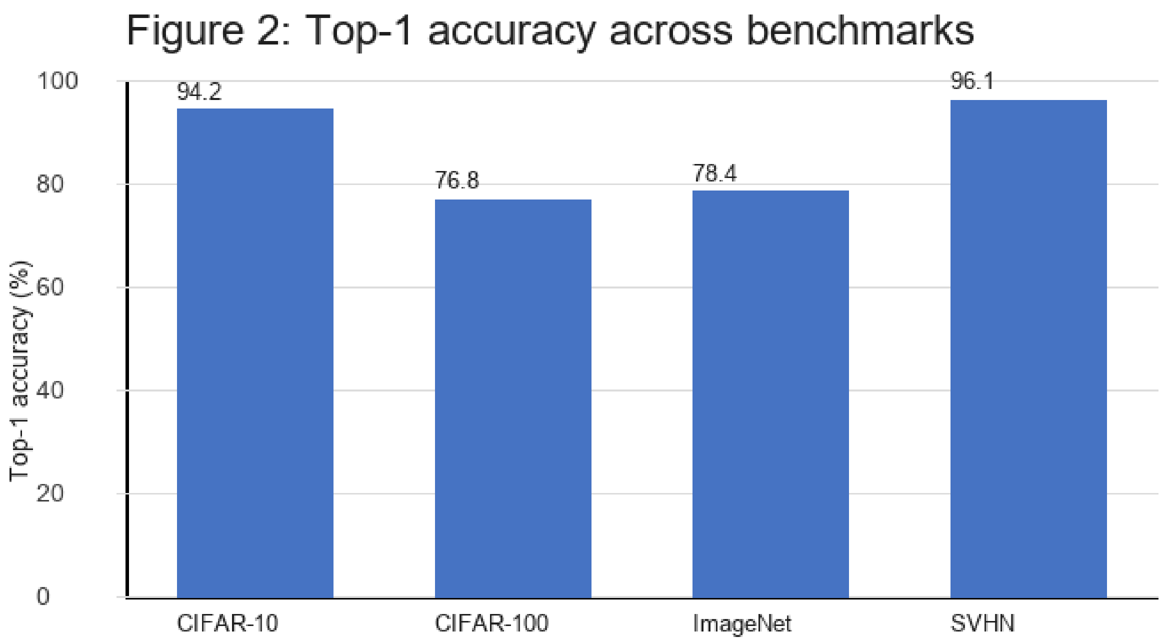

Figure 2: Top-1 accuracy per benchmark. All values from Table 1.

4. Discussion

The numbers indicate that Nebula-Net is most competitive on mid-resolution datasets like CIFAR-100 and ImageNet, where its 12.3M parameter budget still leaves room for strong feature diversity. The model is less dominant on SVHN where simpler baselines already saturate the benchmark.

Training loss curves from our main ImageNet run are presented separately in the supplementary. The loss decreases smoothly without instabilities and we observe no divergence events across three random seeds.

5. Conclusion

We described Nebula-Net, a compact image encoder that achieves 78.4% top-1 accuracy on ImageNet with 12.3M parameters. We believe the accuracy/compute trade-off makes it a practical choice for on-device inference.